PROJECT ASTRO

Assistive Systems for Trusted Robotic Operations

Owner Faisal Farouq | Revision R0.1 | 2026-10-23

This package defines the first controlled research prototype and the evidence needed to release each development stage. It connects requirements, architecture, caregiver workflows, a demonstration script, mass and financial budgets, feature priorities, risk controls, verification, sourcing and the one-year roadmap.

Engineering result: REWORK. Major procurement: BLOCKED. The 55 kg mass figure is an allocation, and the USD 10000 budget is not yet reconciled to technically suitable components. The illustrative commercial-actuation route totals USD 15899. No robot, software-feature or simulation test has been executed, and no fabrication, spending or professional approval is implied.

Control	Baseline
Accountable project owner	Faisal Farouq
Funding constraint	USD 5000-10000; USD 10000 maximum; no increase approved
Mass constraint	<=55 kg target; >55 kg rework; >65 kg major design review
Physical outcomes	Two trials of three tethered steps; separate supported 2 kg handling for 10 repetitions
Evidence state	64 requirements; 26 planned verification protocols; 0 executed system tests
Immediate dependencies	Five owner decisions and named qualified reviewers

How to use this package

The report explains decisions and implementation. The companion workbook holds the complete editable matrices; CSV/JSON and individual Markdown artifacts preserve the project source. Requirement IDs ASTRO-R, verification IDs ASTRO-V, hazard IDs ASTRO-H and decisions D link the records. A document coverage PASS never closes a robot requirement. All provisional quantities must be replaced or accepted through the appropriate engineering gate.

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Appendices contain candidate CAD and simulation specifications, authorization roles, the five next decisions and source references.

1 Mission and scope

ASTRO-A-01 | R0.1 | 2026-10-23 | Owner Faisal Farouq

Define the controlled first-year caregiver-support research mission.

The Project ASTRO Caregiver-Support Research Robot will help caregivers with companionship, low-risk engagement, fictional record organization and communication. Faisal Farouq remains the accountable project owner. Caregivers retain authority, responsibility and clinical judgment. The robot will identify itself as a research support system and will never claim professional-care or emergency-response authority.

Year 1 is a controlled research demonstration. The target robot is 1.829 m tall and no more than 55 kg in operating configuration. A projection above 55 kg requires rework; above 65 kg requires a major design review. The lower body has 10 active joints. The goal is three consecutive slow tethered steps in each of two separate trials, targeting 0.10-0.15 m/s inside a cleared controlled test area with rated robot fall protection. The financial ceiling is USD 10000; feasibility at either USD 5000 or USD 10000 remains unproven.

The companion experience will be developed first on a laptop using fictional profiles, simulated observations, games, medication-response logging, audit history, approved-recipient summaries and telepresence handoff. Every simulated reading retains its value, units, source and time. Reports require exact-content preview and explicit send confirmation. Only approved test recipients may receive them. Endpoint receipt never implies human review.

A separate handling demonstration will move a 2 kg inert object through an approved controlled trajectory for 10 supervised repetitions while the robot is stationary and mechanically secured. Walking with the test object is excluded. The handling module is included in onboard mass accounting; the test object is additional only in the stationary configuration.

Appearance follows the written fallback: removable rigid white rounded plastic panels, visible narrow seams, charcoal protected joint gaps, subtle teal, simple connected eyes and broad flat rounded-rectangle feet. Motion clearance, cooling, service access, sensing, stop access and arrest attachments take priority over cosmetic proportions.

Excluded are real patient data, live EHR integration, diagnosis, medical image interpretation, treatment or dose recommendations, dispensing or medication administration, person lifting/support/restraint, unsupervised high-energy motion and unapproved public operation. A 5 kg payload, untethered gait and clinical use are later-phase questions. No physical performance, fabrication, approval or clinical validation is claimed by this baseline.

Task execution record

Required inputs: Owner brief SRC-01; fallback appearance description; the five owner decisions.

Known values and provisional assumptions: 1.829 m target, <=55 kg target, >65 kg major review, 10 lower-body joints, USD 5000-10000 funding range; no concept image available.

Unknowns: Funding, facility, reviewers, equipment access, selected actuators and approved recipient.

Calculations: Three-step distance example: 3 x 0.15 m = 0.45 m. At 0.10-0.15 m/s this interval takes 4.5-3.0 s; stride and timing are provisional.

Output file: requirements/01_mission_and_scope.md

Implementation steps: 1. Review scope and exclusions. 2. Record owner dispositions D-01 to D-05. 3. Baseline accepted requirements at G0. 4. Begin software-only work.

Acceptance test: One-page mission contains all measurable year-one outcomes and exclusions; owner acceptance is separately recorded.

Stop conditions: Conflicting scope, real patient data, unapproved spending or motion.

Result: PASS - document coverage only; owner acceptance pending

Next dependency: G0 owner dispositions

Physical authorization and supervision: No physical work authorized.

Budget effect: USD 0 committed; cap unchanged.

2 System architecture and boundaries

ASTRO-A-02 | R0.1 | 2026-10-23 | Owner Faisal Farouq

Define modular interfaces that prevent AI failures from becoming unrestricted motion or silent care-data changes.

The mechanical load path runs from the removable exterior and upper structures through the pelvis, lower-body joints and feet to the floor. Exterior panels are non-structural unless later drawings explicitly qualify them. The object-handling

module has a separate support path and enable condition. Robot arrest and handling fixtures stay effective without controller or battery power.

The companion model receives only consented fictional context. It proposes typed actions to a deterministic interaction gateway. That gateway can request a curated game, draft a note, request a report preview or request a named bounded motion skill. It cannot open a motor bus, write a database, choose a recipient, change limits or write original medication instructions. Human confirmations bind to the actual proposed revision or transmission. AI-generated text is untrusted until schema and authorization checks pass.

The care-record and medication-log services own validated transactions. The record service appends corrections with superseded-version links. Audit events record create, view, change, deny and transmit actions. A deterministic summary builder cites source IDs. The model may rephrase a draft only where the final facts are checked against those IDs. Medication instructions remain an immutable original string per human-authorized version.

The communication service receives an approved immutable report snapshot, recipient ID and approval token bound to its hash. A local outbox owns idempotency and retry states. A mock receiver returns endpoint receipt. Human review is a separate explicit authenticated action. Telepresence requires capture consent and a visible call state. External EHR routes and credentials are absent from the Year 1 deployment.

The main motion controller owns standing, weight-shift and gait skills. A separate safety controller supervises independently available enable, stop, watchdog and relevant fault signals. E-stop hardware inhibits hazardous drive enable through a reviewed independent path. A diagnostic controller does not become safety-rated by being independent. Required integrity, channels, monitored contacts, brake behavior and fault coverage need qualified review. Torque-off alone does not hold up a biped. Mechanical support and arrest remain mandatory. Reset clears a latch only after fault resolution and never resumes motion automatically.

Modules M01 structure, M02 lower actuation, M03 feet/contact, M04 protected power, M05 safety, M06 main compute, M07 perception, M08 companion, M09 records, M10 medication, M11 communication, M12 handling, M13 body, M14 evidence and M15 simulation have distinct ownership of state. Evidence subscribes read-only to timestamped events. Simulation emits a distinctly labeled simulation channel and has no hardware credentials. Safety event logging must survive AI or network failure.

The candidate joint tree has l_hip_roll, l_hip_pitch, l_knee_pitch, l_ankle_pitch, l_ankle_roll and the corresponding r_ names. With x forward, y left and z up, roll axes are x and pitch axes are y. There is no hip yaw. This supports investigation of straight-line tethered stepping; turning is excluded. Candidate limits and dimensions live in the CAD and simulation specifications and are not operational safety limits.

Interface summary

Producer	Consumer	Allowed interface
Companion	Typed gateway	Proposed draft or prevalidated skill name only
Human confirmation	Record/medication service	Authorized append or revision; preserve original
Approved preview	Communication outbox	Exact content hash, recipient and version
Motion controller	Joint drives	Bounded reviewed trajectory subject to independent enable
Stops and reviewed faults	Independent safety chain	Latched inhibit; reset does not restart
All services	Evidence service	Read-only synchronized events and versioned logs

Task execution record

Required inputs: Mission, requirements, hazard log, candidate joint topology, data policy and selected safety components later.
Known values and provisional assumptions: 15 modules from the brief; local-first fictional care service; independent safety chain; topology A-02 remains provisional.
Unknowns: Safety integrity requirement, braking strategy, final bus/voltage, model provider and stop-time budget.

Calculations: Stop-envelope screening: d = v*t + v^2/(2*a). For v=0.15 m/s, t=0.10 s and a=0.50 m/s2, d=0.0375 m. This translation-only example excludes limb sweep, fall, latency variation and arrest dynamics; it sets no separation distance.

Output file: architecture/02_system_architecture.md

Implementation steps: 1. Assign interfaces below. 2. Separate credentials and motor networks. 3. Define signed limits and fail states. 4. Freeze topology at G2. 5. Test denied paths before hardware integration.

Acceptance test: ASTRO-V-04, 08, 09, 16 and 19 demonstrate interface consistency and denied unauthorized paths; none is executed yet.

Stop conditions: AI direct bus access, AI writable safety limits, uncontrolled gravity on stopping, or record mutation without authorization.

Result: REWORK - conceptual interfaces defined, hardware and safety response unresolved

Next dependency: G2 safety requirements and architecture freeze

Physical authorization and supervision: Qualified controls/electrical/rigging review before any powered implementation.

Budget effect: No commitments; interfaces may change component costs at G3.

3 Requirements and phase allocation

ASTRO-A-03 | R0.1 | 2026-10-23 | Owner Faisal Farouq

Create testable uniquely identified requirements separated by Year 1, later phases and exclusions.

The authoritative requirement records are requirements/requirements.csv and the Requirements workbook tab. Phase values are Year 1, Later and Excluded. Excluded rows are enforceable negative requirements; they still require verification. Later rows define a gate for future work and do not expand the first-year commitment.

Height acceptance of 1.829 m +/-0.020 m is a proposed build tolerance for owner acceptance, not an alteration of the target. Mass <=55 kg remains a target. Crossing 55 kg triggers rework even though the major-review threshold is >65 kg. Speed is reported as a target-achievement result alongside step completion; a slower safe trial cannot be described as meeting the speed target.

Requirement status BLOCKED - NOT TESTED means implementation or evidence is missing. It does not mean the documentation is incomplete or that failure has been observed. A planned test, plausible calculation or budget allocation cannot close a physical requirement. Each release must preserve the accepted requirement revision, actual configuration and authorized reviewer disposition.

Phase	Count	Evidence
Year 1	50	All untested
Later	6	All untested
Excluded	8	All untested

Complete row-level requirements are in the companion workbook and requirements/requirements.csv. All 64 map to one of the 26 planned protocols.

Task execution record

Required inputs: Owner brief, architecture and test IDs.

Known values and provisional assumptions: 64 requirements; every row maps to a planned verification ID and evidence status.

Unknowns: Owner acceptance, approved tolerances and specialized implementation limits.

Calculations: Coverage = mapped requirements / total requirements = 64/64 = 100 percent planned mapping; executed verification coverage = 0/64 = 0 percent.

Output file: requirements/03_requirement_matrix.md

Implementation steps: 1. Review each shall statement. 2. Accept or revise phase. 3. Freeze accepted revision. 4. Add new IDs for new scope; never recycle IDs. 5. Link executed runs without overwriting failures.

Acceptance test: No duplicate IDs, missing phase or missing verification mapping; no unexecuted performance PASS.

Stop conditions: Excluded capability is enabled, a safety limit is guessed, or a requirement loses source/evidence traceability.

Result: PASS - specification mapping complete; all requirements unvalidated

Next dependency: G0 owner acceptance then detailed verification planning

Physical authorization and supervision: Document review only.
Budget effect: No change; scope additions require cost/mass impact review.

4 Caregiver workflows

ASTRO-A-04 | R0.1 | 2026-10-23 | Owner Faisal Farouq

Specify observable consent-led workflows for engagement, medication, observations and staff communication.

Start each session with the research role and SIMULATED banner. Select a fictional profile and identify the actor role. Ask separately for preference memory, note storage, microphone/camera use and report transmission. A user can decline an activity or withdraw capture consent. If free text appears to contain real personal/health information, pause entry and ask the operator to replace it with fictional fixture data; heuristic detection alone is not a privacy guarantee.

Companionship flow: offer a consented game, story, music/relaxation or feelings check-in; accept a selection through speech, touch or communication board. Ask for clarification when uncertain. A sensitive concern yields a short supportive response and an offer to contact the approved caregiver. Record a note only after the actor confirms the wording. Child preset uses age-appropriate curated content and guardian-controlled settings; older-adult preset offers slower speech and repetition; rehabilitation preset uses seated verbal/visual activities unless a separate motion script is approved; limited-speech/mobility preset provides large targets, captions and a manual board. No preset authorizes clinical or person-contact motion.

Medication flow: a deterministic reminder displays the exact original instructions for fictional Demo Medication A, instruction version and source. Ask taken, skipped, postponed or no response. Record it as a patient statement, never confirmed ingestion. Postponed does not change the prescribed schedule and the robot supplies no catch-up or double-dose guidance. A scripted skipped/missed response prepares a report. Show recipient and complete contents, then require explicit send confirmation. No confirmation means no outbound message; retain the pending draft. Human-authorized record changes create a new version and preserve the original.

Observation flow: receive only an allowed simulator fixture or actor-entered simulated value. Validate metric type, units, source, observed time and actor; record both observed and recorded time. Show trends as descriptive values, with missing/stale flags and no diagnostic label. A deliberately artificial threshold crossing says: 'This simulated reading crossed the demonstration review rule. I cannot determine what it means. Would you like to contact your approved caregiver?' Show the exact trigger. The fixture uses an intentionally non-clinical test threshold and is never a clinical limit.

Staff communication flow: choose routine summary, caregiver attention requested, immediate human contact requested or system assistance required. Assemble facts, actor statements, robot observations and unknowns as separate fields. Preview exact recipient, time window, simulated readings, medication responses, concerns and robot status. Approval binds to report hash, version and recipient. Altered drafts return to preview. Send only to an allowlisted mock receiver; audit every attempt and use the same idempotency key on retry. Show queued, attempted, endpoint received and separately human reviewed. Timeout displays unconfirmed delivery and a manual call option. No state implies emergency dispatch.

Corrections append a superseding record with reason, actor, before/after version and audit event. Unauthorized reads and changes are denied and audited. Local mock mappings use Observation-like value/time/source records, Provenance-like origin/revision records and AuditEvent-like access events. These borrow structure from HL7 R4 references SRC-06 to 08; they do not claim FHIR conformance or connectivity.

Task execution record

Required inputs: Fictional profile fixtures, consent scope, approved recipient, deterministic policies and record schema.
Known values and provisional assumptions: All data simulated; four communication priorities are non-clinical; five interaction presets.
Unknowns: Approved recipient identity, retention period, final accessibility review and endpoint details.
Calculations: Retry example: initial attempt at 0 s, retries at 5 s, 15 s and 35 s; after 3 retries show delivery unconfirmed and offer manual contact. Intervals are software test values, not an emergency-response commitment.
Output file: caregiver_workflows/04_caregiver_workflows.md
Implementation steps: 1. Load allowed fixture. 2. Confirm consent and mode. 3. Run the four flows below. 4. Exercise corrections/declines/offline behavior. 5. Record audit and source IDs.
Acceptance test: ASTRO-V-10 to 19 and 21; each decline and failure has an auditable truthful outcome.
Stop conditions: Real data, recipient outside allowlist, clinical advice, approval mismatch, capture consent withdrawn or corrupted audit.
Result: PASS - workflow specification ready for software implementation; no execution claimed

Next dependency: G1 fixture and recipient policy approval
Physical authorization and supervision: Scripted adult actors on laptop; participant work needs an approved protocol.
Budget effect: Within B-06 and B-12 allowances; no spend committed.

5 First year demonstration script

ASTRO-A-05 | R0.1 | 2026-10-23 | Owner Faisal Farouq

Define a reproducible demonstration with separate software, walking and stationary-handling evidence.

Preflight: record run ID, requirement revision, software/model/configuration hashes, operator names and actual date/time. Confirm all data are fictional. Load profile SIM-001 and local test recipient RECIPIENT-UNASSIGNED only as a non-sendable placeholder until the owner names a consenting recipient. Demonstrate the manual human-contact button. State that this is a research prototype and no patient or emergency service is involved.

Segment A on laptop: choose the older-adult preset; show large text/captions and privacy control. Play one short word game; start a feelings check-in and confirm an actor note. In separate scripted branches show the child, adult, rehabilitation and limited-speech/mobility presets. Insert simulated heart rate, oxygen saturation, temperature, activity/rest and patient-reported entries. Show trend, source, unit and time, then correct a deliberately inaccurate fictional entry and show both versions and access history.

Segment B on laptop: display fictional medication instructions exactly as entered; record a skipped dose and decline one send. Show that no message left the outbox. Create a new exact-preview approval and send to the approved mock recipient. Demonstrate endpoint receipt and separately a human review action. Disable the endpoint, show bounded retries and unconfirmed delivery, and offer manual contact. Cross an artificial simulation review rule and show non-diagnostic explanation. Start and end an approved audio/video test call with visible capture state. Export the human-readable shift report and audit trail.

Segment C in the cleared test area after all motion gates: operator verifies support rig, stop function, interlock, thermal baseline, charge state and log synchronization. With no test payload attached, perform one approved three-step sequence, settle, stop and inspect. Reset, then perform the second sequence. Count distinct alternating swing/landing events using foot contact and video. Record pelvis displacement and time, peak/RMS joint loads, temperature, current and tether load. No arrest catch, external propulsion or operator stabilization counts as a successful gait trial. Tether pretension/support limits must be accepted before the run; report any sustained weight assistance. Target speed is assessed on the defined first-toe-off to third stable landing interval with its distance convention fixed in the protocol. Startup/settling timing must be reported separately.

Segment D in a separately secured stationary fixture: disable walking and verify support/payload interlocks. Use a captured 2.00 kg inert object over a guarded catch tray. Proposed motion is 0.10 m vertical travel at <=0.02 m/s and <=0.30 m horizontal reach, with reviewed self-weight, grip and force limits. Execute 10 complete up/dwell/down/dwell cycles. Record each repetition and every stop. No walking occurs with the object. A stopped repetition is retained as an attempt; do not erase it or count it as complete. The full test is rerun only after a reviewed cause and disposition.

Closeout: remove drive enable, secure gravity loads, export raw logs, hashes, configuration, event video and approved-recipient receipt. Record each criterion as PASS, REWORK or BLOCKED with evidence. State measured outcomes and remaining limitations aloud. If physical gates are unavailable, present the software segment and simulation visualization with a clear partial-completion label; the Year 1 physical goals remain unmet.

Task execution record

Required inputs: Accepted test plan, passed prior gates, fictional fixtures, approved recipient and calibrated logging.
Known values and provisional assumptions: Two three-step trials; 10 supported 2 kg repetitions; software first; no public hazardous first integration.
Unknowns: Actual trial speed, step geometry, tether-load criterion, stop limits and facility configuration.
Calculations: Handling example: 0.10 m lift at 0.02 m/s takes 5 s upward; down takes 5 s; 5 s top and bottom dwell gives 20 s/cycle, 200 s for 10 cycles. Proposed timing only.
Output file: tests/05_demonstration_script.md
Implementation steps: 1. Execute software segment. 2. Review physical permits. 3. Execute walking independently. 4. Reset and mechanically secure for handling. 5. Export evidence and limitations.
Acceptance test: All segment criteria below met with complete raw logs; any omitted segment means partial demonstration only.

Stop conditions: Any unreviewed gate, barrier entry, arrest catch, unexpected motion, limit trip, communication misstatement, payload release or loss of synchronized evidence.

Result: REWORK - script defined; physical and software prerequisites absent

Next dependency: G1 software demonstration then G5 to G8

Physical authorization and supervision: All powered work requires named test director, authorized operator, stop operator and qualified support/rigging review.

Budget effect: No commitments; separate segments avoid added integration scope.

6 Mass budget and engineering screens

ASTRO-A-06 | R0.1 | 2026-10-23 | Owner Faisal Farouq

Allocate onboard mass and expose sizing conflicts before selecting actuators.

The component allocations are in budget/mass_budget.csv. They are design ceilings, not a prediction of the as-built robot. The 14 kg lower-body actuation allocation includes ten modules, transmissions and dedicated actuation mounts. Generic structural brackets belong to the structure line; the later BOM must prevent double counting. External robot arrest, test fixture, floor hardware and the inert payload are outside operating robot mass, but are separately listed and rated. A stationary configuration with the 2 kg object has 57 kg moving/retained robot-plus-object mass at the 55 kg robot target. Walking with that payload is prohibited. Removing a module requires a separately weighed and versioned configuration.

Mass growth: 47.8 x 1.20 =57.36 kg before any unused reserve is counted. This exceeds the 55 kg target by 2.36 kg. At 40 percent growth, 47.8 x 1.40 =66.92 kg, exceeding the 65 kg major-review threshold by 1.92 kg. Growth reserve is headroom, not extra hardware and must not be added again to these grown component sums.

Panels: for assumed area A=2.5 m2, thickness t=0.0015 m and density rho=1050 kg/m3, m=A*t*rho=3.9375 kg. Adding 1.2625 kg for panel supports/overlap gives 5.2 kg. Doubling to 0.003 m produces 9.1375 kg, a 3.9375 kg increase. Area, plastic density, process thickness and required impact/fire performance need material selection and actual CAD; large solid-printed shells can exceed this budget.

Gravity screen with g=9.81 m/s2 and m=55 kg: W=539.55 N. A horizontal COM offset e=0.10 m from a loaded ankle gives tau=W*e=53.96 N m. At e=0.15 m, tau=80.93 N m. Multiplying by an illustrative dynamic screening factor of 1.5 yields 80.93 and 121.40 N m. This is not a final factor of safety or joint load envelope. It omits link inertia, impacts, compliance and load sharing. A full-body inverse-dynamics trajectory and fault-load cases are required.

For illustration, a 35 kg effective upper load at a 0.15 m hip lever gives 35*9.81*0.15=51.50 N m static; a 0.20 m knee lever gives 68.67 N m. The same 1.5 screen gives 77.25 and 103.01 N m. The effective mass and lever arms are assumptions. The manufacturer's AK80-64 reference lists 48 N m rated and 120 N m peak (SRC-02). Even the 0.10 m ankle static example exceeds that rated value. Low walking speed does not remove sustained gravitational torque or winding heating. The 120 N m peak value is not a continuous holding rating.

Foot screen: a 0.32 x 0.22 m rectangle has 0.0704 m2 area. A 55 kg single-support load uniformly spread over it gives 7.66 kPa, but real contact area and local pressure differ. The ideal lateral half-width is 0.11 m; an illustrative 0.025 m edge allowance leaves 0.085 m at the same coordinate origin. Dynamic ZMP, friction and contact uncertainties still require simulation and tests; this is no stability proof. Estimated COM height from provisional component centers is computed as sum(m_i*z_i)/sum(m_i); the reserve has no physical location and is excluded until assigned.

Handling screen: a 2 kg object at horizontal reach 0.30 m contributes 2*9.81*0.30=5.886 N m; at 5 kg it contributes 14.715 N m, 2.5 times as much, before module self-weight, motion, friction and safety factors. A counterbalance or independent supported module reduces the biped integration burden. It does not remove the need for structural/retention review.

Energy screen: assume ten joints each average 40 N m at 0.50 rad/s simultaneously. P_mech=10*40*0.50=200 W. At illustrative efficiency 0.60 plus 60 W electronics, P_bus=200/0.60+60=393.33 W. A 48 V, 8 Ah nominal pack stores 384 Wh=1.3824 MJ. At 80 percent usable energy, ideal runtime is 307.2/393.33=0.781 h=2812 s. This is not a runtime prediction; holding losses, peak loads, temperature, converters, aging and duty cycle are unresolved. Average bus current for this example is 8.19 A. Manufacturer ADC/phase current must not be summed as battery DC current. Peak bus, fuse, conductor, regen and contactor sizing require a reviewed electrical model.

Thermal work remains open: compute tau_RMS=sqrt(sum(tau_i^2*dt_i)/sum(dt_i)), convert motor-side torque to current using vendor gearbox efficiency and torque constant conventions, calculate copper/driver loss, and model temperature with measured thermal parameters. No winding resistance convention or peak-duration limit has been assumed sufficient for

purchase. Robot arrest sizing likewise cannot use W alone: a simple energy screen F_avg approximately m*g*(1+h/d) excludes peak amplification and rig geometry and is intentionally not used to select rated equipment.

Subsystem	Allocation kg
Primary structure and load-bearing joint brackets	12.0
Ten lower-body actuators and transmissions	14.0
Feet soles and ground-contact sensing	2.5
Battery pack with enclosure and BMS	3.5
Protected power distribution and converters	1.4
Independent safety controls and stops	0.6
Computer audio and local display	0.9
Perception sensors	0.6
Supported object-handling module	3.0
Passive upper limbs and head supports	2.0
Rigid exterior panels and panel mounts	5.2
Harnesses connectors and cable protection	1.3
Fasteners seals and labels	0.8
Component allocation	47.8
Unassigned growth reserve	7.2
Target allocation total	55.0

Task execution record

Required inputs: Provisional geometry and material densities; component datasheets and CAD masses later.

Known values and provisional assumptions: 47.8 kg allocated components plus 7.2 kg growth reserve =55.0 kg target; payload and external fixtures accounted separately.

Unknowns: Actual masses, COM, inertias, structural stresses, full duty cycle, foot friction and arrest loads.

Calculations: All calculation examples below and workbook Engineering/Mass tabs; none is a validated selection or physical result.

Output file: budget/06_mass_budget.md

Implementation steps: 1. Replace allowances with datasheet/CAD values. 2. Compute COM/inertias by configuration. 3. Run inverse dynamics and thermal duty analysis. 4. Weigh assemblies. 5. Reconcile remaining reserve at every gate.

Acceptance test: ASTRO-V-02/03; no duplicated/omitted components, all onboard parts counted, measured <=55 kg target, independent engineering sizing review.

Stop conditions: Projected >55 kg requires rework; >65 kg major review; missing load cases or peak-only actuator selection blocks procurement.

Result: REWORK - allocation and screening calculations only

Next dependency: G2 CAD mass/inertia model and load cases

Physical authorization and supervision: No fabrication or loading authorized; calculations require independent review.

Budget effect: Mass growth requires a cost revision; no monetary change accepted.

7 Financial budget and feasibility

ASTRO-A-07 | R0.1 | 2026-10-23 | Owner Faisal Farouq

Preserve the funding cap while exposing unquoted allowances and commercial actuator cost conflict.

The ceiling budget deliberately totals USD 10000. The actuator line is an affordability constraint, not evidence that appropriate ten-joint actuation is available for USD 3000. All remaining lines are unquoted allowances. B-09 includes rated robot arrest and anchor access; B-11 includes limited independent review; B-13 includes freight, tax, duties and standards access; B-14 includes spares and contingency. Reducing safety or omitting review to make the total fit is not a viable budget solution.

The manufacturer's page displayed USD 889.90 as a starting price when checked for this baseline (SRC-02). A driver option is present, so the exact with-driver variant, lead time, warranty, compatible bus and landed price must be confirmed in writing. Ten modules at that displayed minimum cost USD 8899, consuming 88.99 percent of the USD 10000 cap and leaving USD 1101 for the remaining USD 7000 allowance. The illustrative total is USD 15899. This is a feasibility reference with optimistic non-actuator allowances, not a selected BOM, purchase recommendation, firm quote or demonstrated minimum project cost. Actuator suitability remains unproven by the torque and thermal screens.

At the USD 5000 lower funding case, the current USD 7000 non-actuator allowance alone exceeds funding by USD 2000. There is no credible full first-year build plan at USD 5000 on current evidence. Software-first demonstration and engineering/quotation work can proceed as no-spend tasks using equipment only after its availability is confirmed. Completion of software alone must be recorded as a partial project outcome.

Accounting assumption A-07 excludes paid owner/development labor, major facility construction and clinical/regulatory deployment. It includes allowances for qualified fabrication, fixture/tool access and independent reviews, whose sufficiency is unknown. If the funding limit must include paid engineering labor or unavailable facilities, add those costs and reopen feasibility. Record donated/reused equipment with condition, replacement value, ownership and associated cash costs; a donation does not waive safety qualification.

Budget revision rule: the owner accepts every baseline change through a decision record showing previous total, new total, difference, cause, phase impact and funding source. No such increase is approved in R0.1. Existing committed spend is recorded as USD 0 in this package because no purchases were made by this work; prior project spending, if any, is unknown and must be supplied before financial release.

Budget line	Ceiling allowance USD
Ten lower-body actuator assemblies	3,000
Structure and qualified fabrication	650
Feet and contact sensors	180
Battery charger and power distribution	650
Independent safety hardware	450
Compute perception audio and display	500
Plastic panels and mounts	550
Supported handling module	350
Rated robot fall protection and anchorage access	1,000
Fixtures tools and inspection equipment access	500
Qualified independent engineering review	600
Software communication and content licenses	120
Freight taxes duties and standards access	650
Spares and uncommitted contingency	800

Budget line	Ceiling allowance USD
Total ceiling allocation	10,000
Commercial-reference total	15,899
Reference excess over cap	5,899

Task execution record

Required inputs: USD 10000 cap, budget allowances, official manufacturer price reference and later landed quotes.

Known values and provisional assumptions: USD 10000 ceiling allocation; USD 7000 non-actuation total; USD 3000 left for ten actuator assemblies.

Unknowns: Landed prices, safety equipment rental, tax/duty, review hours, reused equipment ownership and paid labor.

Calculations: Affordable actuator allowance=(10000-7000)/10=USD 300. Example reference=10*889.90=8899; total=7000+8899=15899; over cap=5899 or 58.99 percent.

Output file: budget/07_financial_budget.md

Implementation steps: 1. Confirm accounting scope. 2. Request quotes without purchasing. 3. Replace allowances and reconcile reserve. 4. Seek compliant design/reuse/sponsorship decisions. 5. Release exact BOM only at G3.

Acceptance test: All required cash costs including safety, freight/taxes and reserve covered within cap with written quotes; no unsupported free-resource assumptions.

Stop conditions: Total forecast >10000 USD, quote/configuration mismatch, omitted safety cost, or donor equipment/access not documented.

Result: BLOCKED - major procurement is not financially reconciled

Next dependency: D-01 then G3 landed BOM and feasible actuation route

Physical authorization and supervision: No procurement, fabrication or load testing authorized.

Budget effect: Cap stays USD 10000; reference estimate exceeds it by USD 5899; no increase approved.

8 Ranked feature backlog

ASTRO-A-08 | R0.1 | 2026-10-23 | Owner Faisal Farouq

Rank useful additions without silently expanding the first-year scope.

The complete ranked backlog is requirements/feature_backlog.csv and the Backlog workbook tab. Benefit 1 means low expected caregiver value and 5 high. Risk 1 means low incremental risk and 5 high; cost and development time use the same low-to-high ordinal scale. The score supports discussion; mandatory safety, privacy and accessibility controls are implemented as requirements even if a numerical rank changes. Ties use stable feature IDs. Changing a score in the workbook recomputes score and numerical rank; sort the table by rank after editing, because row order is a baseline snapshot.

Consent, accessibility, human-contact controls, summaries, daily check-ins, privacy mode, curated engagement, maintenance reminders and approved telepresence are first-year software work. Basic appointment reminders and static wayfinding use fictional schedules and do not imply public navigation. The manual board provides a non-speech fallback.

Teach-back is limited to exact approved material and cannot certify understanding. Multilingual interaction needs reviewed translations. Family media requires consent and moderation. Room control requires approved interfaces and bounded permissions. Delivery, laundry and meal requests are communication relays before physical transport. Rehabilitation games require qualified clinician-authored motions and a separate motion release. Cart/trolley/cargo alternatives are evaluated for heavier loads before biped payload growth. A counterbalanced or detachable supported module is a Year 1 design option rather than an extra promised robot arm.

Feature	Score	Scope
Consent dashboard and persistent simulated labels	25	Year 1 core
Large text high contrast captions and speech rate	25	Year 1 core

Feature	Score	Scope
Manual human-contact button and communication board	24	Year 1 core
Caregiver shift summary and explanation of actions	23	Year 1 core
Battery maintenance and cleaning reminders	23	Year 1 core
Daily feelings check-in and comfort preferences	22	Year 1 core
Privacy mode with visible microphone and camera off	22	Year 1 core
Games stories licensed music and relaxation	21	Year 1 core
Approved telepresence call	21	Year 1 core
Appointment reminders and static wayfinding	20	Year 1 core

Task execution record

Required inputs: Capability brief, dependency readiness and ordinal benefit/risk/cost/time judgments.

Known values and provisional assumptions: 22 features scored on benefit, risk, cost and time from 1 to 5; dependencies can block any rank.

Unknowns: User-study benefit, actual effort, license costs, endpoint access and qualified reviewers.

Calculations: Score=2*benefit+(6-risk)+(6-cost)+(6-time), maximum 25. Example consent dashboard=2*5+5+5+5=25. Score is an editorial planning heuristic, not quantified safety or ROI.

Output file: requirements/08_feature_backlog.md

Implementation steps: 1. Review score definitions. 2. Resolve dependencies before selection. 3. Implement Year 1 core. 4. Re-score additions after evidence. 5. Approve scope changes through new requirements.

Acceptance test: All ideas requested in brief represented or grouped explicitly; ranking reproducible; no later feature presented as committed.

Stop conditions: New feature changes clinical scope, real-data policy, physical risk or cap without review.

Result: PASS - ranked planning backlog; benefits and effort unvalidated

Next dependency: G1 core software backlog selection

Physical authorization and supervision: Physical additions need their own reviewed motion/fixture gates.

Budget effect: No spend approved; relative cost ratings are not price quotes.

9 Hazard and ethical risk register

ASTRO-A-09 | R0.1 | 2026-10-23 | Owner Faisal Farouq

Identify credible harm, risk controls and evidence required before release.

Severity uses 1 negligible harm, 2 minor reversible injury or inconvenience, 3 serious injury or substantial privacy harm, and 4 fatal/permanent injury or potentially severe care-related misuse. These are conservative consequence categories for screening, not a quantified clinical risk analysis. Likelihood is UNASSESSED for every record. P0 means the relevant release is blocked until the control is verified and reviewed; P1 means resolve before the relevant demonstration stage. Neither category estimates probability.

The complete register is risk_records/hazards.csv and the Hazards workbook tab. Faisal Farouq is accountable for disposition and appointments, not presumed qualified to sign every specialized design. Reviewer fields contain roles marked unassigned; they do not invent professional sign-off. Residual risk remains UNASSESSED and acceptance NOT RECORDED throughout R0.1.

Stop design needs explicit gravity behavior. Removing motor power can cause collapse. The physical safety case must show how the arrest/support retains the robot and object without battery, compute, network or a held torque command. A safety relay, contactor or independent microcontroller alone does not establish the integrity of the whole function. The safety reviewer must define and validate response time, fault detection, reset behavior, limits and retained-load behavior.

ISO 13482 and ISO 13850 official abstracts are design-review references (SRC-03/04). No paid full-standard assessment, compliance claim, certification or regulatory determination is made. Recheck applicable editions and jurisdiction at the architecture gate. Friendly styling increases the need for clear boundaries because spectators may approach or touch the robot. Year 1 physical demonstrations use barriers and scripted adult operators; they are not patient trials. No contact-force number has been borrowed from a different robot standard to imply that this full-size biped is safe for contact.

Risk grouping	Relevant IDs	Release implication
Falls and motion	H-01 to 05, 11, 14, 17, 20	Physical release blocked until qualified control review
Care and privacy	H-06 to 10, 12, 16	Scripted fictional use and denial/consent tests required
Service and evidence	H-13, 15, 18, 19	Maintenance, honest evidence and cost/gate discipline

Task execution record

Required inputs: Requirements, architecture, test configuration, qualified risk review and operator procedures.

Known values and provisional assumptions: 20 hazard/ethical records; consequence severity 1-4; probability and residual risk unknown.

Unknowns: Exposure, actual failure rates, required safety integrity, validated human contact limits and residual acceptance.

Calculations: No risk-priority multiplication is used: unknown probability cannot support a numeric residual-risk reduction or a safe claim.

Output file: risk_records/09_hazard_and_ethical_risk_register.md

Implementation steps: 1. Review hazards by configuration. 2. Add site-specific hazards. 3. Assign named qualified reviewers. 4. Verify controls by linked tests. 5. Record residual risk and release disposition.

Acceptance test: Each record has harm, priority, controls, requirement/test links, accountable owner and a reviewer role; residual acceptance stays pending until evidence.

Stop conditions: Any P0 release control absent, unknown stop response, unqualified arrest design, real patient data or unsupported clinical claim.

Result: REWORK - preliminary register; residual risks not assessed or accepted

Next dependency: G2 qualified hazard review and site survey

Physical authorization and supervision: All physical activity needs a test permit and qualified supervision; this register is not that permit.

Budget effect: Safety controls remain inside the financial model; cost inadequacy blocks release.

10 Verification and evidence matrix

ASTRO-A-10 | R0.1 | 2026-10-23 | Owner Faisal Farouq

Connect every requirement to a reproducible acceptance method and honest evidence status.

The verification matrix in tests/verification_matrix.csv and the Verification workbook tab includes method, pass criterion, prerequisites and the reason each test is blocked. The requirement list in each verification record is generated from the requirement-to-test mapping to avoid stale cross references. Excluded functions have denial tests. Later-phase items have configuration and claim inspections rather than invented execution results.

Before executing a physical protocol, add numeric trip limits, sampling rates, equipment/fixture ratings, calibration validity, test location, operators, exclusion geometry, emergency procedure and an independently approved stop-time budget. Those values are intentionally not guessed in this baseline. Absence of any required value blocks enable. A provisional geometry angle is not an approved software limit.

Evidence record schema: evidence_id; test_id; requirement_ids; actual date/time and time-zone/UTC; actor/operator; reviewer; run result; requirement revision; hardware configuration; software and model hashes; signed limit-set hash; calibration references; environment; raw-log paths and SHA-256 hashes; deviations; stop events; limitations and disposition. Never prepopulate a future run as passed. Document-only QA records describe authoring consistency and rendering, and are explicitly outside robot verification.

Gait evidence must disclose tether loading. A visually plausible animation, simulation with ideal actuators, externally supported stepping motion or footage without synchronized logs cannot establish the full physical goal. Simulator

mass/inertia/friction sweeps and failure injections support engineering decisions but do not validate the actual prototype. Handling and walking get different configuration IDs and test permits.

Protocol group	Tests	Current evidence
Role geometry cost kinematics	V-01 to V-04	None
Walking area shell safety limits	V-05 to V-09	None
Data consent interaction calls records	V-10 to V-17	None
Medication send perception retry	V-18 to V-21	None
Handling interlocks evidence integration future scope	V-22 to V-26	None

Task execution record

Required inputs: Requirement baseline, 26 test specifications, qualified approved limits and test equipment later.

Known values and provisional assumptions: All 64 requirements mapped; 26 verification protocols planned; zero system verification runs executed.

Unknowns: Calibration records, actual software/hardware, run results, environmental conditions and reviewers.

Calculations: Planned traceability 64/64=100 percent; executed system verification 0/26=0 percent. Authoring QA is separate and cannot improve either physical result.

Output file: tests/10_verification_matrix.md

Implementation steps: 1. Approve protocol and limits. 2. Record configuration/calibration. 3. Run only authorized stage. 4. Capture failures and raw logs. 5. Review result. 6. Invalidate affected evidence when configuration changes.

Acceptance test: A PASS has a unique run, exact requirement/config revision, calibrated evidence and reviewer disposition; no pending test is represented as passed.

Stop conditions: Missing prerequisite, invalid calibration, lost logs/time sync, altered configuration, safety trip or fabricated evidence.

Result: REWORK - verification plan complete at baseline level; all system evidence missing

Next dependency: G1 software tests and G2 detailed physical test procedures

Physical authorization and supervision: Planned procedures do not authorize hazardous execution.

Budget effect: Test facility, review and instrument access remain unquoted allowances.

11 Make versus buy and procurement plan

ASTRO-A-11 | R0.1 | 2026-10-23 | Owner Faisal Farouq

Prepare sourcing and fabrication decisions without buying from nominal or peak torque alone.

The ten sourcing decisions are in procurement/make_vs_buy.csv and the Make Buy workbook tab. Buy protected commercial energy equipment, sensors and suitable safety components. Use qualified fabrication for structural load paths and fixtures. Make the deterministic record, consent, audit, summary and communication services. Reuse compute only after availability, ownership and suitability are documented.

Actuator request for information: exact model/hardware/firmware and driver option; output continuous torque versus speed; peak torque duration and recovery; allowable duty cycle versus ambient and cooling; motor and gear thermal limits; efficiency map; rotor/output inertia; backlash/compliance; encoder behavior on reboot; power-off/backdrive and brake behavior; radial/axial/moment bearing loads; mechanical life; absolute and full-charge voltage range; bus/phase current definitions; regen handling; fault/enable interface and safety claims; CAN protocol; dimensional STEP and 2D toleranced interfaces; service lead time; complete landed quote and warranty/returns. Without these, an impressive peak torque does not qualify a joint.

Every purchase request includes accepted requirement IDs, intended joints/configuration, drawing revision, vendor evidence, planned incoming inspection, cost impact, unresolved deviations and independent reviewer disposition. G3 release covers specific items and amounts. A prototype actuator order is still a purchase and needs separate owner authorization plus a safe bench-test plan; this baseline places none.

Incoming inspection records part/revision/serial number, physical condition, dimensions, weight, electrical ratings, connector/polarity and documentation. Qualified personnel execute energized checks on a restrained fixture after power/safety acceptance. Nonconforming material is quarantined and excluded from motion configurations. Fabrication drawings must include material, finish, critical tolerances, fastener specifications, load cases, inspection and revision. The master CAD envelope is not a released fabrication drawing.

Subsystem	Sourcing route	Gate
Lower-body actuator assemblies	Buy commercial candidate; make adapters only after sizing	G3
Load-bearing skeleton and fixtures	Qualified shop fabricates released CAD; prototype envelopes in low-energy materials	G4
Fall protection anchors and support rig	Buy/rent engineered rated equipment with qualified rigging service	G2
Battery charger and protected distribution	Buy matched protected pack/charger; qualified electrical assembly	G3
Safety chain	Buy suitable safety components; reviewed integration	G3
Feet sensing and compliant sole	Buy sensors; make structural foot and replaceable sole	G3
Panels and cosmetic parts	Make removable lightweight shells via low-cost commercial process	G4
Compute perception audio display	Reuse verified equipment or buy commercial	G1
Care software records and communication	Make deterministic services; buy licensed model/telepresence service only if needed	G1
Supported handling	Buy counterbalance/retention components; fabricate qualified fixture or detachable module	G3

Task execution record

Required inputs: Frozen interfaces, load/duty models, quotes, safety requirements, budget and named fabricator.

Known values and provisional assumptions: Ten sourcing packages defined; no supplier selected or order placed; commercial actuator is a reference only.

Unknowns: Vendor curves, thermal conditions, delivery/returns, braking, mounting loads, complete quotes and service access.

Calculations: Commercial reference mass=10*0.85=8.5 kg before secondary transmission/mounts; 14 kg allocation leaves 5.5 kg for those parts, but gives no suitability proof.

Output file: procurement/11_make_versus_buy.md

Implementation steps: 1. Issue information/quote checklist. 2. Compare technical compliance. 3. Reconcile full BOM. 4. Complete independent G3 review. 5. Sign exact purchase lines. 6. Inspect delivered revision before use.

Acceptance test: Every major item has data, interface drawing, duty suitability, landed cost, inspection method and named release authority.

Stop conditions: Peak-torque-only selection, absent driver configuration, missing thermal data, cap overrun, unreviewed structural drawings or unnamed fabricator.

Result: BLOCKED - quote packages defined; design and procurement prerequisites absent

Next dependency: G2 architecture freeze then G3 procurement release

Physical authorization and supervision: Only an explicitly named authorized qualified fabricator may perform released structural work.

Budget effect: No expenditure approved; current reference route exceeds cap.

12 One year roadmap and release gates

ASTRO-A-12 | R0.1 | 2026-10-23 | Owner Faisal Farouq

Sequence six workstreams through explicit procurement fabrication and test gates while preserving a software-first result.

Workstream A maintains requirements, safety/ethical risks, assumptions, decisions, access matrix, verification and excluded claims throughout the year. Faisal Farouq chairs owner decisions; independently qualified people review specialized engineering. A document status PASS is not a physical release.

Workstream B builds the companion and caregiver laptop experience in months 1-3, then tests and refines it through month 12. Complete fictional-data, medication, consent, summary, audit and communication failure tests before connection to any robot. This work can progress while mechanical feasibility remains open, using confirmed available equipment and no unapproved spending.

Workstream C develops the master CAD skeleton, five-joint-per-leg candidate, feet, load cases, mass/COM/inertias, protected power, safety stop behavior and panel schedule in months 2-4. Workstream D develops the pinned-version MuJoCo model concurrently using the same names/axes/limits and documented mass/inertia estimates. Standing, weight-shift and three-step sweeps include friction, delay, backlash, thermal/duty constraints and failure injection. The simulator includes a calibrated tether approximation; its assistance is logged. Nothing here claims those models or simulations have been executed.

Workstream E prepares quotes and interface/inspection packages in months 3-5, with expensive orders blocked until G3. If no credible whole-system route fits USD 10000, the owner must retain the cap and revise resources, design or schedule through a recorded decision. The walking commitment is not silently removed, and a budget increase is not implied. Fabrication starts only at G4 with a named qualified shop and released drawings.

Workstream F follows the exact staged order: 1 software-only caregiver demonstration; 2 sensors; 3 individual actuator module; 4 power system; 5 safety controller and stops; 6 one joint; 7 one leg on fixture; 8 lower-body structure without walking; 9 supported standing; 10 weight shifting; 11 one tethered step; 12 three consecutive tethered steps; 13 repeated three-step sequence; 14 stationary supported handling; 15 shell clearance/thermal testing; 16 full caregiver-support demonstration. Each stage has separate evidence and a release for the next. Simulation evidence before purchasing is distinct from physical powered-actuator acceptance after receipt.

Indicative timing: G0 month 1; G1 months 1-3; G2 months 2-4; G3 months 4-5; G4 months 5-6; G5 months 6-8; G6 months 8-10; G7 months 10-11; G8 month 12. Relative timing starts only when G0 is accepted. Staffing and lead times are unknown, so this is a dependency plan, not a promise of a completed full-size biped in one year.

At G8 archive the accepted requirements, as-built BOM/drawings, actual spend, configuration hashes, calibrated logs, approved protocols, raw failures, verified outcomes and limitations. A software-only fallback may be shown when motion gates are blocked, labeled as partial completion. No public demonstration is the first test of two hazardous subsystems together.

Gate	Timing	Required release evidence
G0 Baseline acceptance	Month 1	Accepted mission, exclusions, budget accounting, five decisions, role appointments, assumption dispositions
G1 Software baseline	Months 1 to 3	Laptop demo, fictional-data boundary, consent, audit, medication and report tests ASTRO-V-10 to 19 and 21
G2 Architecture and facility readiness	Months 2 to 4	Frozen ten-joint CAD interfaces; mass allocation; facility and fall envelope; support concept; hazard dispositions
G3 Procurement release	Months 4 to 5	Accepted requirements; torque-speed/duty/thermal/power models; simulated standing and three-step evidence; vendor data; landed BOM <=10000 USD; interface drawings; named fabricator; inspection plan
G4 Fabrication release	Months 5 to 6	Released drawings; material/load-path calculations; stop/support design; incoming inspection; fabrication process and inspection plan
G5 Fixture and electrical acceptance	Months 6 to 8	Stage 2 sensors; 3 actuator module; 4 power; 5 safety; 6 joint; 7 fixture leg evidence; stop-time and load-retention acceptance
G6 Supported locomotion progression	Months 8 to 10	Stage 8 lower body; 9 standing; 10 weight shift; 11 single step; 12 three steps; 13 repeat; separate passed evidence at each stage

Gate	Timing	Required release evidence
G7 Handling and enclosure acceptance	Months 10 to 11	Stage 14 stationary handling; stage 15 shell clearance and thermal evidence; physical isolation from gait
G8 Integrated research release	Month 12	Stage 16 complete caregiver demo; all verification outcomes reviewed; limitations, costs, configuration and release archive

Task execution record

Required inputs: Accepted baseline, staffing capacity, facility, quotes, CAD/simulation evidence and safety reviews.

Known values and provisional assumptions: Months 1-12 are relative to G0 acceptance; indicative dates 2026-11-02 to 2027-11-01 are provisional.

Unknowns: Staff hours, skills, fabrication lead time, facility access, budget feasibility and actuator lead times.

Calculations: No schedule confidence percentage is calculated without resource/lead-time data. A missed gate shifts dependent work; calendar dates never waive prerequisites.

Output file: revision_history/12_one_year_roadmap.md

Implementation steps: 1. Accept G0 and name roles. 2. Run the six workstreams below. 3. Review gate evidence, cost and mass monthly. 4. Release only next authorized stage. 5. Archive final evidence and limitations at G8.

Acceptance test: G0-G8 have named authority, deliverables and release restrictions; all 16 integration stages stay in order.

Stop conditions: Over-cap BOM, unstaffed specialist role, unapproved gate, missing support or attempted schedule bypass.

Result: REWORK - dependency roadmap defined; dates and resources unconfirmed

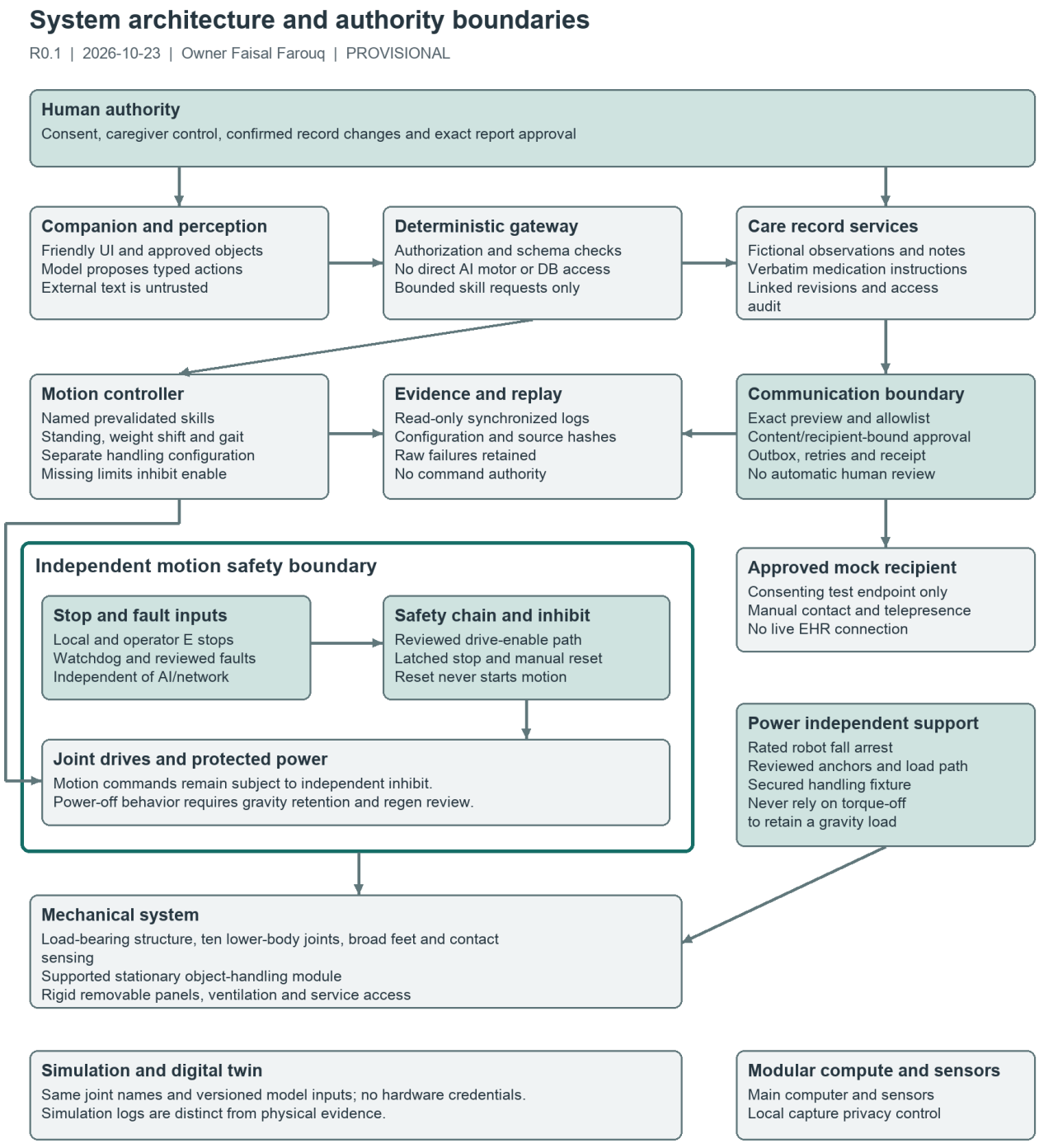
Next dependency: D-01 to D-05, then G0 baseline acceptance

Physical authorization and supervision: Procurement and physical work require gate-specific written authorization; none supplied here.

Budget effect: Cap unchanged; no committed spend; schedule/resource changes require a revised estimate.

Appendix A Engineering implementation specifications

Architecture diagram



The model proposes actions. Deterministic services authorize records and communication. Motion remains subject to an independent safety path and power-independent load retention.

Master CAD specification

REWORK. This directory contains specifications, not fabrication CAD. Candidate dimensions and joint envelopes are provisional. Ten joint centers, bearing axes, hard stops, encoder references, cable bend radii, fastener access, foot support polygons, panel swept volumes, battery/compute masses, COM and principal inertias must be modeled. CAD and simulation must share the same coordinate frame and joint names. Confirm 1.829 m overall height; reconcile the 0.43 m link lengths with ankle/hip offsets and body proportions. Broad feet remain 0.32 x 0.22 m candidate envelopes. No hip yaw. Panel seam dimensions and finger access require a reviewed gap/guarding design; no arbitrary gap is approved here. Output the next released skeleton as master_skeleton_R0_2.step plus dimensioned interface drawings and mass_properties_R0_2.csv. No structural fabrication authorized.

MuJoCo implementation specification

BLOCKED for validation. No MJCF model or simulation run is delivered or claimed in R0.1. Next inputs are accepted topology, per-link mass/inertia tensors, contact/friction assumptions, vendor torque-speed/duty limits and calibrated tether concept. Pin the simulator version and verify force-limit semantics against SRC-05. Model all ten named joints and a free base, explicit foot contact geometry, joint envelopes, actuator saturation/latency, encoder and IMU noise/delay, contact sensing and a unilateral compliant tether approximation with recorded force. Do not use a fixed-base/equality support as if it proved free walking.

Implement standing, weight shift and three-step trajectories separately. Sweep mass/COM uncertainty, sole friction, contact stiffness, backlash, controller delay and actuator limits. Inject sensor freeze/dropout, communications loss, saturation, unexpected contact, support slack and power-loss response. Record seed, simulator version, model/config hash, dt, trajectory, states, contacts, joint position/speed/torque, limit events, power estimate and tether force. Reduce time step until key outcome changes fall within an independently selected convergence tolerance. Compare simple gravity loads to analytical screens, then calibrate against restrained module tests. No simulation PASS transfers automatically to hardware. Stop procurement if required torque/duty is outside vendor envelopes or the success depends on unrealistically helpful tether forces. Next outputs: model_R0_2.xml, controllers, sweep_config_R0_2.json and replayable evidence.

Software implementation plan

REWORK. R0.1 supplies schema/workflow specifications and fictional fixtures, not a working application. Build the laptop UI first with persistent SIMULATED banner and five interaction presets. Use local role accounts and a transactional local store. Separate companion gateway, consent, record, medication, deterministic summary, outbox and audit modules. The model receives no direct DB/motor/network privileges. Add typed APIs for propose_note, confirm_revision, record_simulated_observation, log_medication_response, preview_report, approve_report and send_approved_report. Every mutation validates actor, role, revision and confirmation. The outbox requires recipient allowlist and content-hash-bound approval; the default fixture has no approved recipient and cannot send.

Minimum observation fields: record_id, profile_id, simulated=true, metric, typed value, unit, source_id, actor_id, observed_at, recorded_at, statement_type, version, supersedes_id and correction_reason. Unknown values are null with a reason, never zero. Medication stores immutable original_instruction per authorized version; a response is a statement, not verified ingestion. Audit records create/read/change/transmit/deny with actor, target/version, time and result. Original audit events are append-only at application level; tamper evidence needs protected hashes/export and is not absolute immutability.

Priority enums: ROUTINE, CAREGIVER_ATTENTION, HUMAN_CONTACT_REQUESTED, SYSTEM_ASSISTANCE. Transmission states: DRAFT, APPROVED, QUEUED, ATTEMPTED, ENDPOINT_RECEIVED, DELIVERY_UNCONFIRMED, HUMAN_REVIEWED. HUMAN_REVIEWED requires a separate authenticated test-recipient action. No state represents emergency dispatch. Use a local receiver with idempotency keys and crash-safe retries. Bind all runtime inputs to documented fixtures; no EHR credentials. Next output is software/app with tests for ASTRO-V-10 to 19 and 21, not physical control code.

Candidate joint envelopes

Per-leg joint	Axis	Envelope rad	Status
hip_roll	1 0 0	-0.25 to 0.25	CAD envelope only
hip_pitch	0 1 0	-0.55 to 0.55	CAD envelope only
knee_pitch	0 1 0	0.00 to 1.30	CAD envelope only
ankle_pitch	0 1 0	-0.35 to 0.35	CAD envelope only
ankle_roll	1 0 0	-0.20 to 0.20	CAD envelope only

Use identical l_ and r_ prefixes. Candidate envelopes are not approved operational limits. Missing signed operating limits inhibit motion.

Appendix B Authorization and evidence

Role	Allowed actions	Boundary
Owner Faisal Farouq	Approve scope budgets and reviewer appointments	Cannot substitute for specialist competence; no automatic physical-task authority
Patient test actor	View own fictional profile; choose activity; confirm note and report; request human	No safety-limit or medication-instruction edits
Authorized caregiver test actor	Enter fictional notes and original medication instructions; confirm revisions	No autonomous clinical decisions delegated to robot
Companion model	Read minimized consented fictional context; propose typed drafts and approved skill requests	No motor bus, raw DB writes, recipients, safety limits or transmission credentials
Record and medication services	Validate/commit authorized transactions and append versions	No unaudited mutation; no dosing engine
Communication service	Send exact approved snapshot to allowlisted mock recipient	No unapproved content/recipient; no false review or emergency claims
Qualified test director and operator	Execute only signed test permit and released configuration	Must be named before physical task; cannot waive independent review
Qualified safety and engineering reviewers	Review within documented competence; sign exact scope and limits	Unassigned until appointed; no sign-off implied
Safety controller and stop hardware	Inhibit hazardous motion independently	No conversational dependency; reset does not restart
Evidence service	Append synchronized logs and configuration hashes	No ability to command motion or rewrite original evidence

Assumptions requiring resolution

ID	Subject	Resolve by
A-01	Geometry	G2
A-02	Joint topology	G2

ID	Subject	Resolve by
A-03	Mass	G2
A-04	Power	G3
A-05	Facility	G2
A-06	People	G0
A-07	Budget	G3
A-08	Payload	G2
A-09	Data	G1
A-10	Calendar	G0
A-11	Appearance	G0
A-12	Research protocol	G1

Complete provisional assumptions and consequences are in risk_records/assumptions.csv and the Assumptions workbook tab. No reviewer or authorized fabricator has been named by this package.

Appendix C Five next owner decisions

D-01 Budget and walking feasibility

Retain USD 10000 cap; authorize quotation and no-spend engineering only; require credible whole-system route before actuator commitment.

USD 300 per complete lower-body assembly affordable under current allowances versus USD 889.90 displayed starting-price reference.

Status: BLOCKED pending owner decision. Resolve at G0 to G3. Owner: Faisal Farouq.

D-02 Facilities and accountable qualified people

Name test director, controls/electrical/mechanical reviewers, rigging provider and fabricator; confirm controlled facility and access costs.

No walking or load testing can be released without these roles and a reviewed power-independent support system.

Status: BLOCKED pending appointments. Resolve at G0 to G2. Owner: Faisal Farouq.

D-03 Ten-joint topology and motion scope

Accept the proposed five joints per leg with no hip yaw for straight-line tethered steps, or request architecture rework.

Adding hip yaw while retaining both ankle axes creates 12 lower-body joints; dropping ankle roll changes balance control.

Status: REWORK pending topology decision. Resolve at G2. Owner: Faisal Farouq.

D-04 Handling arrangement and reach

Choose a stationary counterbalanced or detachable supported module; use provisional 0.30 m horizontal reach and 0.10 m vertical travel for analysis.

2 kg at 0.30 m adds 5.89 N m before arm self-weight, acceleration, friction or margin; 5 kg adds 14.72 N m.

Status: REWORK pending handling decision. Resolve at G2. Owner: Faisal Farouq.

D-05 Software scope consent and approved recipient

Name one consenting test recipient and choose local mock communication, five interface presets and fictional fixtures; approve retention/recording policy.

No actual recipient message is authorized by this package; allowlist and per-report confirmation remain required.

Status: REWORK pending workflow decision. Resolve at G1. Owner: Faisal Farouq.

Appendix D Sources and revision record

SRC-01 Owner brief

Pasted text supplied in this chat

Source of project limits; no concept image supplied. Checked 2026-10-23.

SRC-02 CubeMars AK80-64 manufacturer page

https://www.cubemars.com/goods.php?id=1143

Displayed starting price USD 889.90; 48 N m rated, 120 N m peak, 0.85 kg. Driver configuration, duty conditions and landed price require a written quote. Checked 2026-10-23.

SRC-03 ISO 13482 official scope

https://www.iso.org/standard/53820.html

Personal care robot safety reference; medical device robots outside this published scope. Edition applicability and full text review remain open. Checked 2026-10-23.

SRC-04 ISO 13850 official scope

https://www.iso.org/standard/59970.html

Emergency stop design reference; not a brake, fall-arrest or complete functional-safety design. Checked 2026-10-23.

SRC-05 MuJoCo modeling documentation

https://mujoco.readthedocs.io/en/latest/modeling.html

Model force and control clamping; version must be pinned before model implementation. Checked 2026-10-23.

SRC-06 FHIR R4 Observation

https://hl7.org/fhir/R4/observation.html

Reference for local mock observations; no conformance or live interoperability claim. Checked 2026-10-23.

SRC-07 FHIR R4 AuditEvent

https://hl7.org/fhir/R4/auditevent.html

Reference for access and security-event records. Checked 2026-10-23.

SRC-08 FHIR R4 Provenance

https://hl7.org/fhir/R4/provenance.html

Reference for record origin and revision history. Checked 2026-10-23.

R0.1 records only planning and document-authoring checks. No cost increase, professional sign-off, physical test success, fabrication, model simulation or clinical approval was created or inferred. New evidence must identify the actual run, configuration and reviewer. Planned dates remain provisional.